

CLASS TEST

ASSESSMENT TOOL - 1

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COURSE: COMPUTER
VISION.

1. IMAGE ENHANCEMENT IN SPATIAL DOMAIN.

DEFINITION:

* Image enhancement in the spatial domain improves image quality by directly modifying pixel values.

a) Gray - Level Transformations.

Modify pixel intensity values.

Techniques: Negative, Log, Power-law [Gamma], Contrast Stretching.

Use: Improves brightness and contrast.

b) Histogram Processing.

Improves image contrast by redistributing gray levels.

Techniques: Histogram Equalization, Histogram Specification.

Use: Enhances low-contrast images.

c) Spatial Filtering.

Uses a mask [kernel] to process neighboring pixels.

Smoothing Filters: Mean, Gaussian, Median
[noise removal].

Sharpening Filters: Laplacian, Sobel,
unsharp Masking [edge enhancement].

Justification:

Gray-level transformation $\rightarrow$ Brightness
adjustment.

Histogram processing $\rightarrow$ Better contrast.

Spatial filtering $\rightarrow$ Noise removal and
sharpen edges.

Application:

* Medical imaging, Satellite images, CCTV
Surveillance, Document enhancement.

Conclusion:

* Spatial domain enhancement improves
image quality by enhancing brightness,
contrast, reducing noise, and highlighting
important details.

2. IMAGE SMOOTHING IMAGE VS IMAGE SHARPENING.

* Image Smoothing Image Sharpening removes noise enhances edges Blurs image makes clearer uses low-pass filters uses high-pass filters.

* Removes high-frequency components enhances high-frequency components.

Image Smoothing.

* Smoothing reduces noise and unwanted details. It uses low-pass filters.

Working:

* A filter mask is moved over the image.

* Pixel values are calculated using neighboring pixels.

* This reduces sudden intensity changes.

Eg: Mean filter, Median, filter.

Application: Noise removal, medical images preprocessing.

IMAGE SHARPENING:

* Sharpening enhances edges and fine details. It uses high-pass filters.

Working:

- * Detects sudden changes in intensity.
- * enhances edges and boundaries.
- * Makes the image appear clearer.

Eg: Laplacian, Sobel, Prewitt.

CONCLUSION:

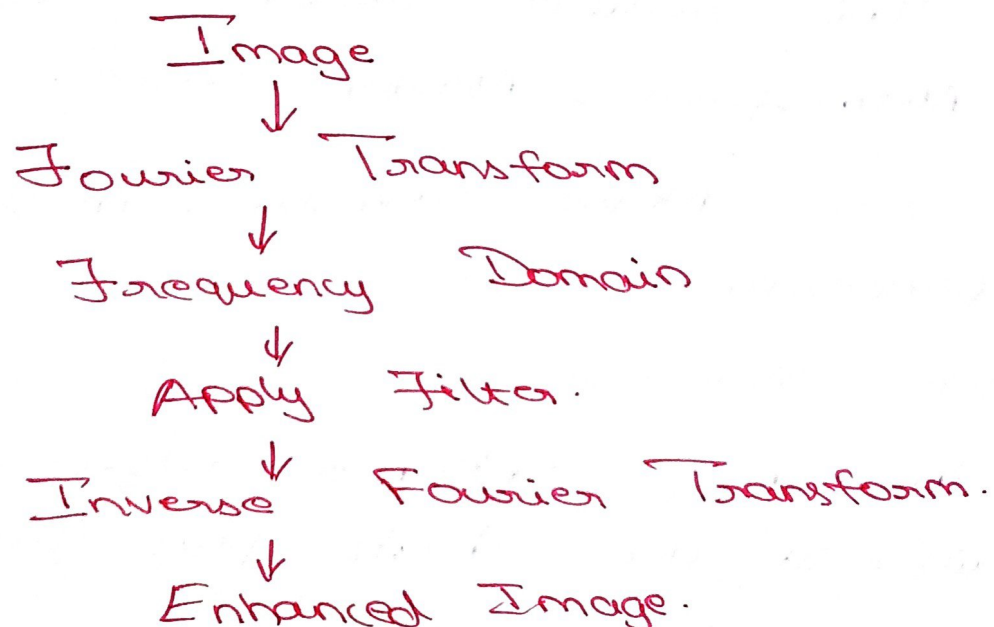
* Smoothing is used when noise needs to be removed, while sharpening is used when edges and details need to be enhanced.

3. IMAGE ENHANCEMENT IN FREQUENCY DOMAIN

INTRODUCTION:

* Frequency - domain enhancement processes an image according to its frequency components.

PROCESS:



Low - Pass FILTER:

* A low-pass filter allows low frequencies and reduces high frequencies.

EFFECTS:

- * Removes noise.
- * Reduces fine details.
- * Smooths the image.
- * Can cause blurring.

APPLICATIONS:

- * Noise removal, image smoothing.

Preprocessing.

HIGH - PASS FILTER:

* A high-pass filter allows high frequencies and reduces low frequencies.

EFFECTS:

- * Enhances edges.
- * Highlights fine details.
- * Sharpens the image.

APPLICATIONS:

- * Edge enhancement, sharpening defect detection.

CONCLUSION:

* Low-pass filters are suitable for smoothing, while high-pass filters are suitable for sharpening & edge enhancement.

4. IMAGE SEGMENTATION:

DEFINITION:

* Image Segmentation is the process of dividing an image into meaningful regions or objects.

TYPES:

1. Point Detection: Detects isolated points
2. Line Detection: Detects lines.
3. Edge detection: Detects object boundaries.
4. Thresholding: Separates objects based on intensity.
5. Region-based: Groups similar neighboring pixels.

SUITABLE TECHNIQUE:

* For object and defect detection, edge detection with thresholding is suitable.

Reason: Edge detection identifies the object's boundary, while thresholding separates the object from the background.

CONCLUSION:

* Segmentation helps in object recognition, medical imaging, defect detection, and image analysis.

5. EDGE DETECTION, EDGE LINKING AND BOUNDARY DETECTION.

INTRODUCTION:

* Edges are important features that represent sudden changes in intensity. Edge detection, linking, and boundary detection are important for recognizing objects and defects.

1. EDGE DETECTION:

* Edge Detection identifies sudden intensity changes in an image.

Eg: Sobel, Prewitt, Roberts, Canny.

Dark Region | Bright Region.
 ↑
 Edge.

USES: Object recognition, Segmentation, defect detection.

2. EDGE LINKING:

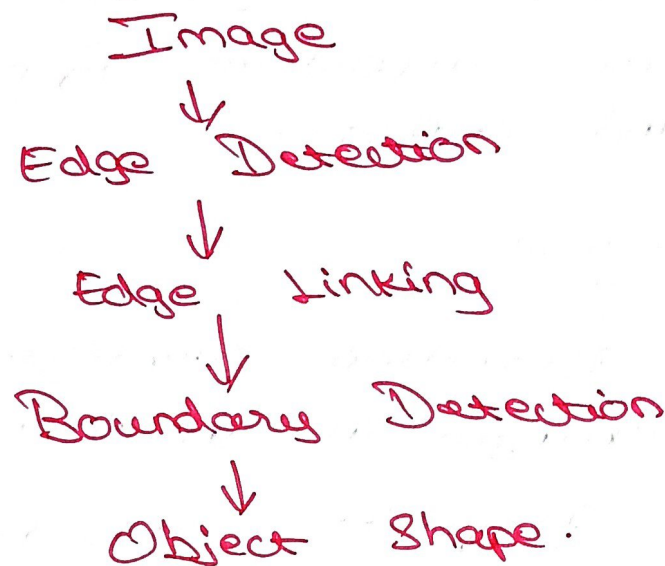
* Edge linking connects broken or disconnected edge pixels.

Broken edges → Edge Linking → Continuous edge.

* It uses neighboring pixels, direction, and intensity to connect edges.

3. BOUNDARY DETECTION:

* Boundary detection identifies the complete outline of an object.



CONCLUSION:

* Edge detection finds edges, edge linking connects them, and boundary detection forms the complete object boundary. Together, they improve object recognition and defect detection.